Do Now · retrieval

Lesson 145 · Packet Switching, Circuit Switching and Reliability

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the job of DNS?

Q2

What is the accumulator for?

Q3

Which is faster in the worst case on sorted arrays, linear or binary search?

Q4

Which logic gate outputs 1 only when both inputs are 1?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Packetise data — Large message split into addressed packets for shared links.

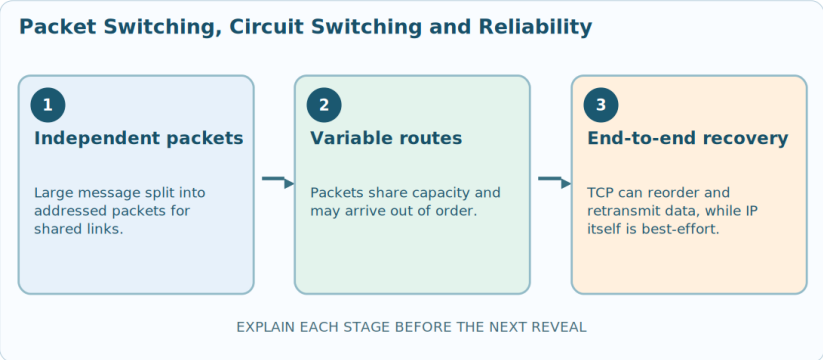
Core idea 2

Route over network — Packets share capacity and may arrive out of order.

Core idea 3

Reassemble reliably — TCP can reorder and retransmit data, while IP itself is best-effort.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

packet switching — Data divided into packets that may take different network routes.

circuit switching — A dedicated end-to-end circuit established before communication.

latency — Delay experienced by data in transit.

retransmission — Sending data again when loss is detected by a reliable protocol.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A 3-packet file crosses a packet-switched network. Packets arrive 2,1,3. Explain handling.

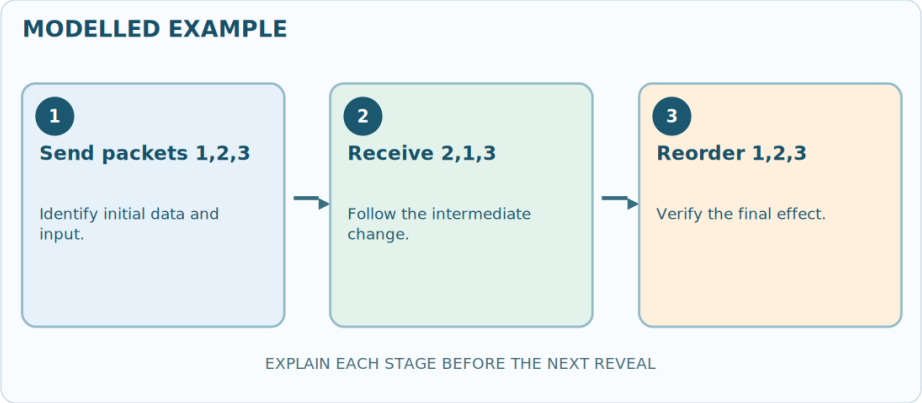
Follow the worked method

1. Send packets 1,2,3 — Identify the relevant information and make a prediction.

2. Receive 2,1,3 — Explain the change using correct technical terms.

3. Reorder 1,2,3 — Packets can follow different routes or encounter different queue delays. With a reliable transport such as TCP, sequence numbers permit correct reordering before delivery of the byte stream; missing data can be retransmitted.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Reserve dedicated circuit
Identify the relevant information and make a prediction.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define packet switching and circuit switching, highlighting two differences. [6]

Question 2

Explain why packets in a message may reach the destination out of order. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate packet switching versus circuit switching for video calls, including reliability and resource efficiency. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

IP alone guarantees that every packet is received in the original order.

Use key vocabulary from memory

1. Explain packet switching in your own words.

2. Explain circuit switching in your own words.

3. Explain latency in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.